Sheilla Mumbi Macharia

Data Scientist | Software Developer

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SUMMARY

Data Science practitioner with a background in Applied Mathematics and hands-on experience across predictive modeling, backend automation, and AI-powered reporting. Skilled in Python, FastAPI, and Streamlit for deriving actionable insights and shipping data-driven applications, from notebook-based analysis to scheduled pipelines and production APIs. Passionate about turning raw data into meaningful stories that drive smarter decisions, with proven success in projects spanning health prediction, agency workflow automation, retail analytics, and agricultural forecasting.

SKILLS

- **Languages & Core:** Python, SQL, HTML/CSS, Git, GitHub
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Jupyter Notebook
- **Backend & Automation:** FastAPI, Flask, PostgreSQL, Jinja2, Gemini API
- **Visualization & BI:** Tableau, Power BI, Streamlit, Plotly, Excel
- **Frontend:** React, Tailwind CSS
- **Core Skills:** Data Cleaning, Feature Engineering, Predictive Modelling, KPI Design, Sentiment & Time-Series Analysis, Business Intelligence

PROJECTS

Tafsiri: AI Client Workflow Automation for Creative Agencies - Link

Python · FastAPI · PostgreSQL · Gemini API · 5-person team project

- Built the Client Payment Risk / Cash Flow Predictability engine as a FastAPI service within a 5-person team building an end-to-end AI agency-workflow tool (discovery calls, proposal generation, quoting, manager approval, and QuickBooks integration).
- Identified and resolved two data leakage issues that were silently inflating model performance.
- Debugged backend infrastructure issues (import-order bugs, Postgres sequence desync) and documented fixes via detailed PRs.

PulseMetrix: Coronary Heart Disease Risk Prediction - Link

Python · Scikit-learn · SMOTE · Streamlit · Plotly

- Built a clinical prediction pipeline to estimate 10-year coronary heart disease risk from Framingham Heart Study data, engineering 11 clinical features such as pulse pressure and mean arterial pressure.
- Selected a regularized Logistic Regression over Gradient Boosting (test ROC-AUC 0.70 vs 0.64) and tuned the decision threshold to 0.30, raising recall from 63% to 91% so high-risk patients are not missed.
- Deployed an interactive Streamlit app with risk gauges, patient-vs-benchmark charts, and plain-language recommendations.

Kenya Regional Crop Yield Prediction - Link | Live App

Python · Scikit-learn · Pandas · NumPy · Matplotlib

- Developed an agricultural forecasting pipeline to predict regional crop yields in Kenya by integrating climate data, pesticide usage, and historical production metrics.
- Consolidated multi-source data from Kaggle, HarvestStat, and OpenAfrica, resolving spatial mismatches and engineering custom seasonal climate features (Long vs. Short Rains).
- Implemented and optimized Random Forest and XGBoost regressors, outperforming the baseline Linear Regression model by capturing complex, non-linear interactions between rainfall and crop output.

- Deployed the final model as a web application via Flask on Render, providing actionable forecasts to support early-warning food security planning and resource allocation.

Supermarket Performance Optimization Pipeline - Link
Python (Pandas, NumPy, Statsmodels) · SQL · Tableau · Power BI

- Engineered an end-to-end retail data science framework to analyze the operational variables driving footprint efficiency and revenue velocity across store locations.
- Conducted bivariate OLS regression analysis to discover a deterministic linear relationship ($\rho = 0.999$) between store area and inventory capacity, alongside identifying a traffic paradox where gross footfall decoupled from total sales volumes.
- Built interactive dashboards in Tableau and Power BI to translate complex retail metrics into actionable stakeholder insights.

EXPERIENCE

Data Scientist Consultant | Genesis Design Factory Limited (Jitume Digital Media Factory)
August 2026 – Present | Nairobi, Kenya

- Develop data science and AI components alongside the Flask/FastAPI backend team.
- Delivered the Client Payment Risk engine for Tafsiri, combining invoice-lateness predictions with Gemini-generated executive briefings and reminder and escalation queues for managers.

EDUCATION

University of Nairobi | Bachelor of Science in Applied Mathematics — Sep 2021 – Sep 2026

Moringa School | Data Science Bootcamp — Aug 2025 – Feb 2026

- Intensive hands-on program in Python, machine learning, and data visualization.
- Capstone projects: SyriaTel Customer Churn Prediction (92% accuracy, 64% churn recall), Kenya Regional Crop Yield Prediction.

CERTIFICATIONS

- Data Science Program, eMobilis Technological Institute
- Python for Data Analysis, FreeCodeCamp
- Machine Learning & Data Visualization, Kaggle

ADDITIONAL INFORMATION

- **Hackathons:** 48-Hour Jitume AI Hackathon, building AI-agent concepts for creative business operations.
- **Other projects:** Data Refinery Studio (Streamlit data-cleaning app) · Twitter Brand Sentiment NLP · Aviation Risk Insights (Tableau)
- **Languages:** English (Fluent) · Swahili (Fluent)